AIMS
Airbus Material Specification

Carbon Fibre Reinforced Epoxy Prepreg Unidirectional Tape/180ºC- curing class

Intermediate modulus fibre Structural material

Material Specification

Published and distributed by :
AIRBUS S.A.S.
ENGINEERING DIRECTORATE
31707 BLAGNAC Cedex
FRANCE

Contents

1 Scope

This Material Specification defines the requirements of intermediate modulus carbon fibre reinforced unidirectional tape prepreg, curing at 180ºC, for aerospace application.
Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

This material is used for manufacturing primary airframe and aerospace structures (sandwich and monolithic). The material is to be delivered according to the relevant ABS. The requirements specified in this Material Specification shall not be used as stress allowables.

3 Normative References

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

ABS 5321	Airbus Standard - Carbon fibre reinforced epoxy prepreg - Unidirectional tape 180ºC curing class intermediate modulus fibre - Structural material
AIMS 05-01-000 Part 1	Airbus Material Specification - Carbon fibre reinforced epoxy prepreg - Technical Specification
AIMS 05-01-000 Part 2	Airbus Material Specification - Qualification Programme - Carbon fibre reinforced epoxy prepreg - Qualification and batch release testing UD tape/180ºC curing class - Structural Material - Technical Specification
AIMS05-01-000 Part 100	Carbon fiber reinforced epoxy prepreg Structural material Acceptance Testing and Storage Life Prolongation
AITM 1.0001	Airbus Test Method - Fibre reinforced plastics - Determination of mechanical degradation due to chemical paint strippers
AITM 1.0002	Airbus Test Method - Fibre reinforced plastics - Determination of in-plane shear properties
AITM 1.0003	Airbus Test Method - Fibre reinforced plastic materials - Determination of the glass transition temperatures
AITM 1.0005	Airbus Test Method - Carbon fibre reinforced plastics - Determination of interlaminar fracture toughness energy - Mode I
AITM 1.0006	Airbus Test Method - Carbon fibre reinforced plastics - Determination of interlaminar fracture toughness energy - Mode II
AITM 1.0007	Airbus Test Method - Fiber reinforced plastics - Determination of plain, open hole and filled hole tensile strength
AITM 1.0008	Airbus Test Method - Fiber reinforced plastics - Determination of plain, open hole and filled hole compression strength
AITM 1.0009	Airbus Test Method - Fibre reinforced plastics - Determination of bearing strength
AITM 1.0010	Airbus Test Method - Fibre reinforced plastics - Determination of compression strength after impact
AITM 3.0008	Airbus Test Method - Determination of the extent of cure by differential scanning calorimetry
ASTM E831	Standard test method for linear thermal expansion of solid materials by thermomechanical analysis
EN 2378	Aerospace series - Fibre reinforced plastics - Determination of water absorption by immersion
EN 2557	Aerospace series - Carbon fibre preimpregnates - Determination of mass per unit area
EN 2558	Aerospace series - Carbon fibre preimpregnates - Determination of the percentage of volatile matter

EN 2559	Aerospace series - Carbon fibre preimpregnates - Determination of the percentage of resin and fibre content and the mass of fibre per unit area
EN 2560	Aerospace series - Carbon fibre preimpregnates - Determinantion of the resin flow
EN 2561	Aerospace series - Carbon fibre reinforced plastics unidirectional laminates - Tensile test paralell to the fibre direction
EN 2563	Aerospace series - Unidirectional laminates - Determination of the apparent interlaminar shear strength
EN 2597	Aerospace series - Unidirectional laminates carbon thermosetting resin - Tensile test perpendicular to the fibre direction
EN 2823	Aerospace series - Fibre reinforced plastics - Determination of the effect of exposure to humid atmosphere on physical and mechanical characteristics
EN 2850	Aerospace series - Carbon fibre thermosetting resin unidirectional laminates compression test parallel to the fibre direction
EN 3615	Aerospace series - Fibre Reinforced plastics. Determination of the conditions of exposure to humid atmosphere and moisture absorption
ISO 10119	Carbon fibres - Determination of density
ISO 1183	Plastics - Mehtods for determining the density and relative density of non-cellular plastics

4 Technical Requirements

4.1 General

Specification values are minimum requirements except where stated otherwise.

4.1.1 Material Description

See table 1.

Table 1: Material description

Characteristic	Requirements	
Material type	UD carbon fiber tape/epoxy resin	
Formulation	Fibre: Intermediate modulus carbon fibre Resin: Epoxy, toughened	
Form and method of production	Prepreg, cured in an autoclave at 180ºC	
Technical Specification	AIMS 05-01-000 Part 1 and Part 2	
Resin content by weight, uncured	Grade A 34% nominal	Grade B 33% nominal, 34% nominal and 35% nominal

4.1.2 Worklife and storage conditions

See table 2.

Table 2: Worklife and storage conditions

No.	Properties	Shop condition	Symbol	Unit	Requirements
1	Storage life at -18ºC	-	-	Months	> 12
2	Storage life at (5±2)ºC	-	-	months	[2]
3	Work life [1] (tack life)	30ºC/75% R.H.	-	days	> 10
		(25±2)ºC/(45±10)% R.H.	-	days	> 10

[1] The prepreg can be exposed an additional minimum of 5 days at the above mentioned shop condition provided that the lay-up is fully bagged and ready for use.
[2] To be reported during qualification

4.1.3 Processing

The curing cycle shall be agreed between the manufacturer and Airbus at the onset of the material qualification programme and shall be used to produce all test specimens for qualification testing. The laminates for test purposes shall be cured in an autoclave. The curing cycle shall comply with the general requirements reported in Table 3.
The agreed curing cycle shall be specified in the relevant IPS and shall be used for the manufacturing of batch release test specimens.

Table 3: Processing conditions

Cure type	Autoclave		
Cure conditions	Heating rate	ºC/min	0.2 - 5.0
	Cure temperature	ºC	180 +10/-5
	Cure time	Min	120 - 240
	Cure pressure	MPa	0.3 – 1.1
		Bar	3 - 11
	Cooling rate	ºC/min	0.2 - 3.5
	Vacuum (abs)	MPa	0.005 - 0.1

Note: The use of an intermediate temperature and low pressure dwell is permissible if agreed between manufacturer and purchaser.

4.2 Specific

4.2.1 Physical-chemical properties

Uncured and cured properties see table 4.

Table 4: Physical properties

No.	Properties	Test methods	Unit	Requirements			
Material Code				AIMS 05-01-002			
				A	B	C	D
Uncured				Grade A	Grade B		
1	Prepreg areal weight	EN 2557C	g/m^2	293±13	412±15	400±12	406±12
2	Fibre areal weight	EN 2559C	g/m^2	194±8	268±10	268±6	268±8
3	Resin density	ISO 1183 Method A	g/cm^3	1.29 ± 0.03			
4	Fibre density	ISO 10119	g/cm^3	1.80 ± 0.05			
5	Volatile content	EN 2558	%	2% maximum			
6	Resin content	EN 2559	%	34±2	35±2	33±1.5	34±1.5
7	Physico-chemical tests	AIMS 05-01-000 part 1 (5.3.5.2)	-	See relevant IPS			
8	Resin flow	EN 2560	%	See relevant IPS			
9	Tack	To be agreed between manufacturer and Airbus	-	-			
Cured							
10	Coefficient of thermal expansion	ASTM E831	ºC^{-1}	To be reported			
11	Water uptake	EN 3615, 70ºC/85% R.H./eq.	%	See relevant IPS			
12	Water uptake	EN 2378, 70ºC/water/eq.	%	See relevant IPS			
13	Extent of cure	AITM 3-0008	%	$\geq$ 90			
Note: The stated requirements are average values.							

4.2.2 Mechanical properties

See table 5 (for Grade A and B materials)
See table 6 (for Grade A material)
See table 7 (for Grade B material)

Table 5: Mechanical properties. Grade A (Code A) and GradeB (Codes B, C and D) materials

No.	Properties	Test method	Unit	Conditioning [2]	Test temp. [°C]	Requirements [1] X	X_{min}
Material Code						AIMS 05-01-002 A/B/C/D	
1	Glass transition temperature, Tgonset	AITM 1.0003	ºC	dry	-	-	150
				wet	-	-	120
2a	Tensile strength 0º	EN 2561 type B	MPa	dry	-55	2600	2400
					RT	2400	2200
					70	2400	2200
				wet	70	2300	2200
					90	2050	2000
2b	Tensile modulus 0º	EN 2561 type B	GPa	dry	-55	170 ± 15	
					RT	170 ± 15	
					70	170 ± 15	
				wet	70	170 ± 15	
					90	170 ± 15	
2c	Poisson Ratio	EN 2561 Type B	GPa	dry	RT	0.3	
				wet	70	0.3	
					90	0.3	
3a	Tensile strength 90º	EN 2597 type B	MPa	dry	-55	35	30
					RT	50	45
				wet	70	25	22
					90	23	20
3b	Tensile Modulus 90º	EN 2597 type B	GPa	dry	-55	9.5 ± 1	
					RT	8.0 ± 1	
				wet	70	7.5 ± 1	
					90	7.5 ± 1	
4a	Compression Strength 0º	EN 2850 type B	MPa	dry	-55	1400	1250
					RT	1475	1350
					70	1350	1250
				wet	70	1150	1025
					90	1000	950
4b	Compression modulus 0º	EN 2850 type B	GPa	dry	-55	140 ± 15	
					RT	140 ± 15	
					70	140 ± 15	
				wet	70	140 ± 15	
					90	140 ± 15	
5a	Compression strength 90º	EN 2850 type B	MPa	dry	RT	210	190
				wet	70	150	140
					90	135	125
5b	Compression modulus 90º	EN 2850 type B	GPa	dry	RT	9 ± 1	
				wet	70	9 ± 1.2	
					90	8 ± 0.5	

(continued)

Table 5: Mechanical properties. Grade A (Code A) and GradeB (Codes B, C and D) materials.

No.	Properties		Test method	Unit	Conditioning 2)	Test temp. [°C]	Requirements 1) X	X_{min}
Material Code							AIMS 05-01-002 A/B/C/D	
6a	In plane shear Strength	Grade A	AITM 1.0002	MPa	dry	-55	85	80
						RT	90	85
						70	85	80
						120	70	68
					wet	RT	80	77
						70	74	70
						90	64	62
						120	52	50
6a	In plane shear Strength	Grade B			dry	-55	68	65
						RT	74	70
						70	70	65
						120	65	60
					wet	RT	77	75
						70	68	65
						90	62	60
						120	49	47
6b	In plane shear modulus		AITM 1.0002	GPa	dry	-55	5.5 ± 1.0	
						RT	4.5 ± 1.0	
						70	3.5 ± 1.0	
						120	3.5 ± 1.0	
					wet	RT	4.5 ± 1.0	
						70	4.0 ± 1.0	
						90	3.5 ± 1.0	
						120	2.0 ± 1.0	
7	Interlaminar Shear Strength 0º		EN 2563	MPa	dry	-55	120	117
						RT	85	82
						70	75	70
						120	59	57
					wet	RT	80	75
						70	52	50
						90	52	50
						120	42	40
8	Compression after impact at 25J (25/50/25)		AITM 1.0010	MPa	dry	RT	235	215
					wet	70	215	195
9	G_{1C}		AITM 1.0005	J/m^2	dry	RT	375	300
					wet	RT	300	250
10	G_{2C}		AITM 1.0006	J/m^2	dry	RT	650	620
					wet	RT	550	500

(continued)

Table 5: Mechanical properties. Grade A (Code A) and GradeB (Codes B, C and D) materials.

No.	Properties	Test method	Unit	Conditioning 2)	Test temp. [°C]	Requirements 1) X	Requirements 1) X_{min}
Material Code						AIMS 05-01-002 A/B/C/D	
11a	Interlaminar Shear Strength 0º (reference)	EN 2563 (3)	MPa	dry	RT	85	80
				wet	70	55	50
11b	Interlaminar Shear Strength (skydrol, 70ºC/1000 h)	EN 2563 (3)	MPa	dry	RT	(5)	(5)
				wet	70	-	-
11c	Interlaminar Shear Strength (skydrol / water, 70ºC/1000 h)	EN 2563 (3)	MPa	dry	RT	(5)	(5)
				wet	70	-	-
11d	Interlaminar Shear Strength (fuel, RT/1000 h)	EN 2563 (3)	MPa	dry	70	(5)	(5)
				wet	70	-	-
11e	Interlaminar Shear Strength (MEK, RT/1h)	EN 2563 (3)	MPa	dry	RT	(5)	(5)
				wet	70	-	-
12a	Resistance to strippers (Phenolic)	AITM 1.0001 (4)	MPa	dry	RT	(5)	(5)
12b	Resistance to strippers (Non Phenolic)	AITM 1.0001 (4)		dry	RT	(5)	(5)
				wet	70	-	-
13a	Plain tensile strength	AITM 1.0007A (25/50/25)	MPa	dry	RT	745	700
				wet	70	745	700
13b	Open hole tensile strength (6)	AITM 1.0007B (25/50/25)	MPa	dry	RT	500	475
				wet	70	460	440
					90	460	440
13c	Plain tensile strength	AITM 1.0007A (10/80/10)	MPa	dry	RT	-	-
				wet	70	450	430
13d	Open hole tensile strength (6)	AITM 1.0007B (10/80/10)	MPa	dry	-55	310	295
					RT	300	295
				wet	70	280	270
14a	Plain compression strength	AITM 1.0008A (25/50/25)	MPa	dry	RT	475	430
				wet	70	350	325
14b	Open hole compression strength (6)	AITM 1.0008B (25/50/25)	MPa	dry	RT	275	270
				wet	70	235	220
14c	Plain compression strength	AITM 1.0008A (10/80/10)	MPa	dry	RT	-	-
				wet	70	280	250
14d	Open hole compression strength (6)	AITM 1.0008B (10/80/10)	MPa	dry	-55	285	280
					RT	235	230
				wet	70	195	185
15a	Filled hole tensile strength (6)	AITM 1.0007C (25/50/25)	MPa	dry	RT	430	410
				wet	70	400	360
					120	410	400
16a	Filled hole compression strength (6)	AITM 1.0008C (25/50/25)	MPa	dry	RT	315	300
				wet	70	280	230
					120	260	230
16b	Filled hole compression strength (6)	AITM 1.008C (10/80/10)	MPa	dry	RT	-	-
				wet	70	265	255

(continued)

Table 5: Mechanical properties. Grade A (Code A) and GradeB (Codes B, C and D) materials. (concluded)

No.	Properties	Test method	Unit	Conditioning [2)]	Test temp. [°C]	Requirements [1)] X	Requirements [1)] Xmin
Material Code						**AIMS 05-01-002 A/B/C/D**	
17a	Bearing Strength 2%	AITM 1.0009 (25/50/25)	MPa	dry	RT	620	550
				wet	70	520	470
					90	490	440
					120	470	420
17a	Bearing Strength Ultimate		MPa	dry	RT	790	770
				wet	70	700	670
					90	690	665
					120	600	575
17b	Bearing Strength 2%	AITM 1.0009 (10/80/10)	MPa	dry	-55	630	580
					RT	620	540
				wet	70	530	480
17b	Bearing Strength Ultimate		MPa	dry	-55	1020	1000
					RT	920	900
				wet	70	750	730
17c	Bearing Strength 2%	AITM 1.0009 (50/40/10) (40/40/20)	MPa	dry	-55	700	650
					RT	640	570
				wet	70	450	400
					120	340	310
17c	Bearing Strength Ultimate		MPa	dry	-55	920	880
					RT	850	750
				wet	70	670	640
					120	590	570

1) Requirements for Tension 0º, Compression 0º, Plain, Open Hole and Filled Hole Tension and Compression have been calculated using a theoretical thickness per ply of:
0.184 mm for Grade A materials and 0.250 mm for Grade B materials
2) Wet means 70ºC/85% R.H. according to EN 2823
3) According to AIMS 05-01-000 Part 1 para. 4.2.3.3
4) According to AIMS 05-01-000 Part 1 para. 4.2.3.4
5) The values after conditioning will not have a decrease greater than the corresponding one with the ILSS or in plane shear reference value at equivalent test temperature
6) Requirements calculated with gross section

Table 6: Mechanical Properties. Grade A (Code A) materials (Multidirectional Directed properties)

No.	Properties	Test method	Unit	Conditioning [2]	Test temp. [°C]	Requirements [1] X	Requirements [1] Xmin
Material Code						AIMS05-01-002A	
13e	Plain Tensile Strength	AITM 1.0007A (50/40/10)	MPa	dry	-55	1350	1320
				dry	RT	1350	1320
				dry	70	1450	1420
				wet	70	1350	1320
13f	Open hole tensile strength [3]	AITM 1.0007B (50/40/10)	MPa	dry	-55	760	710
				dry	RT	840	800
				wet	70	840	800
14e	Plain compression strength	AITM 1.0008A (50/40/10)	MPa	dry	-55	820	740
				dry	RT	640	630
				wet	70	530	500
14f	Open hole compresión strength [3]	AITM 1.0008B (50/40/10)	MPa	dry	-55	415	405
				dry	RT	340	330
				wet	70	270	260
15c	Filled hole tensile strength [3]	AITM 1.0007C (50/40/10)	MPa	dry	-55	-	-
				dry	RT	600	575
				dry	70	630	590
				wet	70	590	550
16c	Filled hole compression strength [3]	AITM 1.0008C (50/40/10)	MPa	dry	RT	390	375
				wet	70	360	350

[1] Requirements have been calculated using a theoretical thickness per ply of: 0.250 mm
[2] Wet means 70ºC/85% R.H. according to EN 2823
[3] Requirements calculated with gross section

Table 7: Mechanical Properties. Grade B (Codes B,C and D) materials (Multidirectional Directed Properties)

No.	Properties	Test method	Unit	Conditioning [2]	Test temp. [°C]	Requirements [1]	
						X	Xmin
Material Code						AIMS 05-01-002 B/C/D	
13e	Plain Tensile Strength	AITM 1.0007A (40/40/20)	MPa	dry	-55	1110	1040
					RT	1200	1130
					70	1210	1175
				wet	70	1100	1050
13f	Open hole tensile strength [3]	AITM 1.0007B (40/40/20)	MPa	dry	-55	630	600
					RT	660	640
				wet	70	700	680
14e	Plain compression strength	AITM 1.0008A (40/40/20)	MPa	dry	-55	600	575
					RT	560	520
					70	500	450
				wet	70	450	400
14f	Open hole compresión strength [3]	AITM 1.0008B (40/40/20)	MPa	dry	-55	415	405
					RT	340	320
				wet	70	270	260
15c	Filled hole tensile strength [3]	AITM 1.0007C (40/40/20)	MPa	dry	RT	534	520
					70	570	545
				wet	70	540	520
16c	Filled hole compression strength [3]	AITM 1.0008C (40/40/20)	MPa	dry	RT	400	380
					70	375	360
				wet	70	350	320

[1] Requirements have been calculated using a theoretical thickness per ply of: 0.250 mm
[2] Wet means 70ºC/85% R.H. according to EN 2823
[3] Requirements calculated with gross section

5 Designation

Refer to ABS 5321

6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS05-01-000 Part 1 and Part 2.

7 Incoming inspection

Materials qualified against this AIMS shall undergo “Technical Incoming Inspection” as per AIMS05-01-000 Part 100 prior to use.

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 11/99		New standard
2 11/03		Complete Revision
3 09/05	Table 4-5	Revision of requirements due to availability of test data
4 03/08	Table 1 Table 4 Tables 5-6	Resin content added New material code added New heads, including material Codes A, B, C and D
5 07/13	Chapter 3 Chapter 7	AIMS05-01-000 part 100 added Chapter 7 defined in line with Incoming Inspection (Part 100) implementation